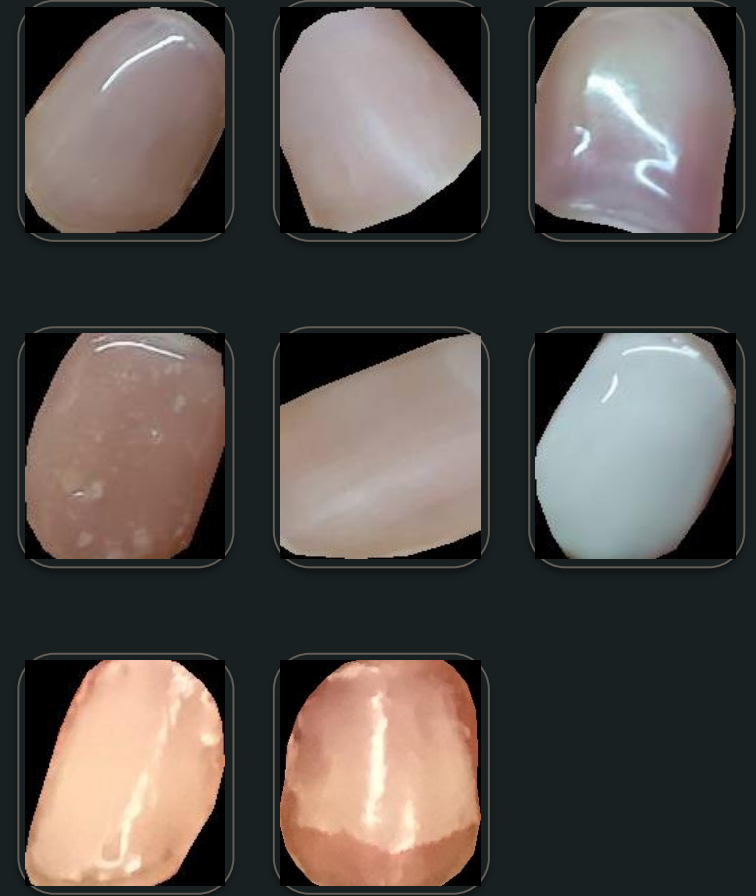


Turning “unusable” nail images into a condition signal

AI & Arts Final Project ·
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Project claim

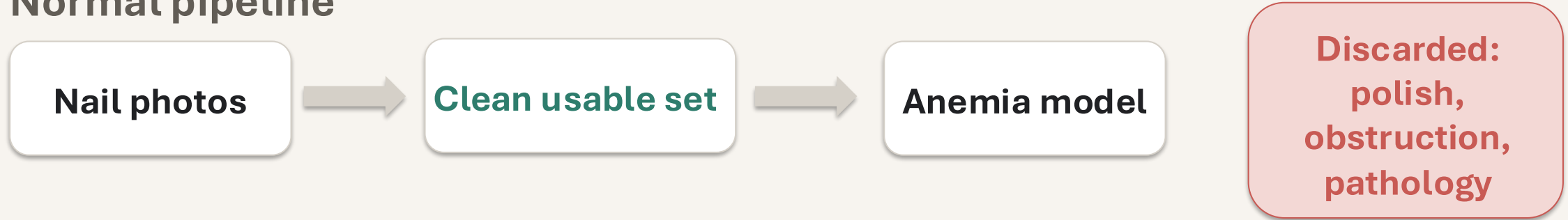
Excluded nail images can become a
learned warning signal
for real-world anemia prediction.



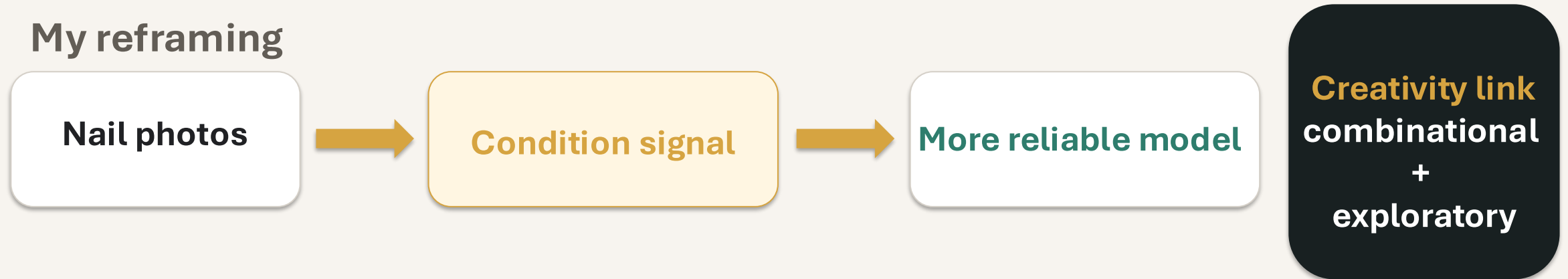
Stop treating excluded images as empty noise.

Proposal idea: rethink “bad data” as structured edge cases that can expand the learning space.

Normal pipeline



My reframing

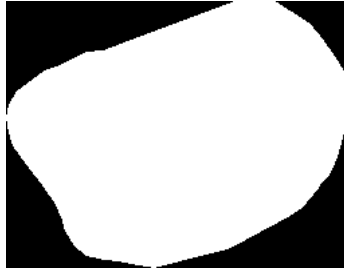


Made the image condition visible by isolating the nail region.

Description: AnyLabeling polygons → nail-only PNG crops → polish/no-polish labels → ResNet18 classifier.



manual polygon label



segmented region



polish example



nail-focused crop

1

draw polygon

2

crop nail only

3

split by group

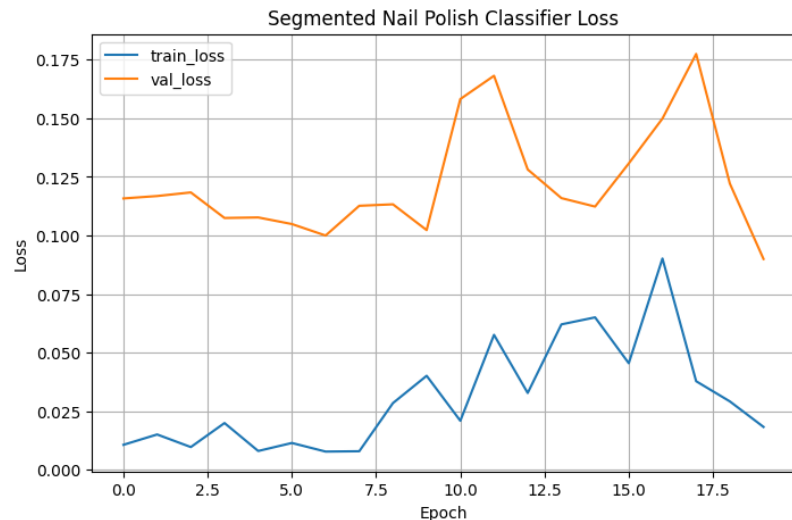
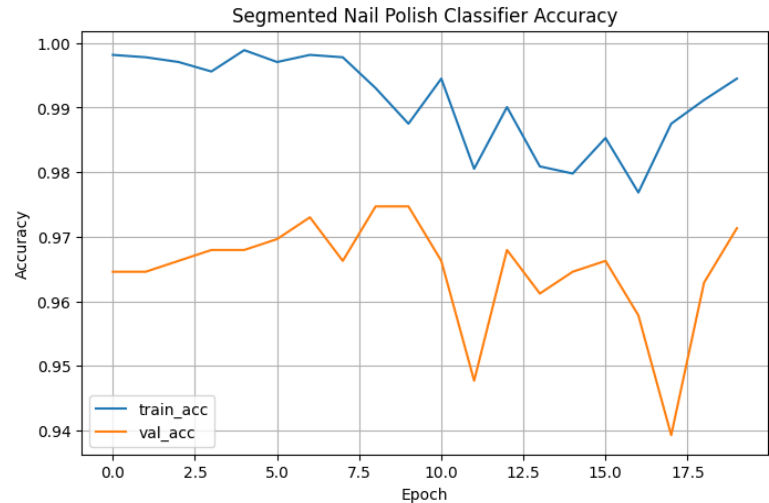
4

fine-tune ResNet18

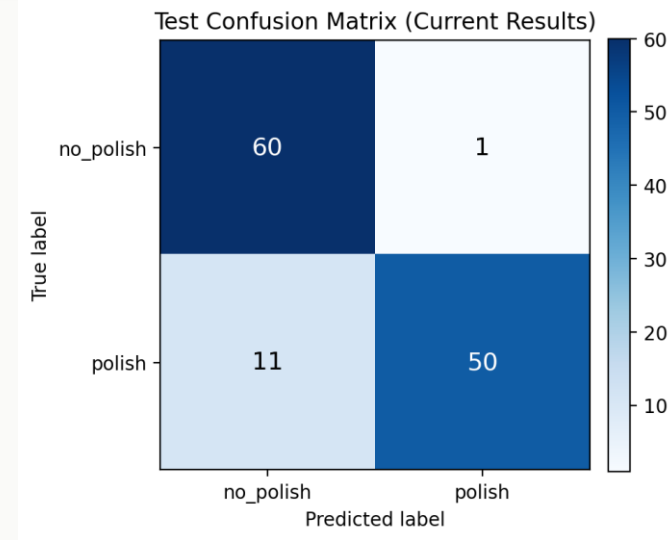
train layer4 + fc

Balanced testing shows a useful polish signal, but polish recall is the weak point.

Training evidence



90.2% test accuracy
macro-F1 = 0.90 · weighted-F1 = 0.90

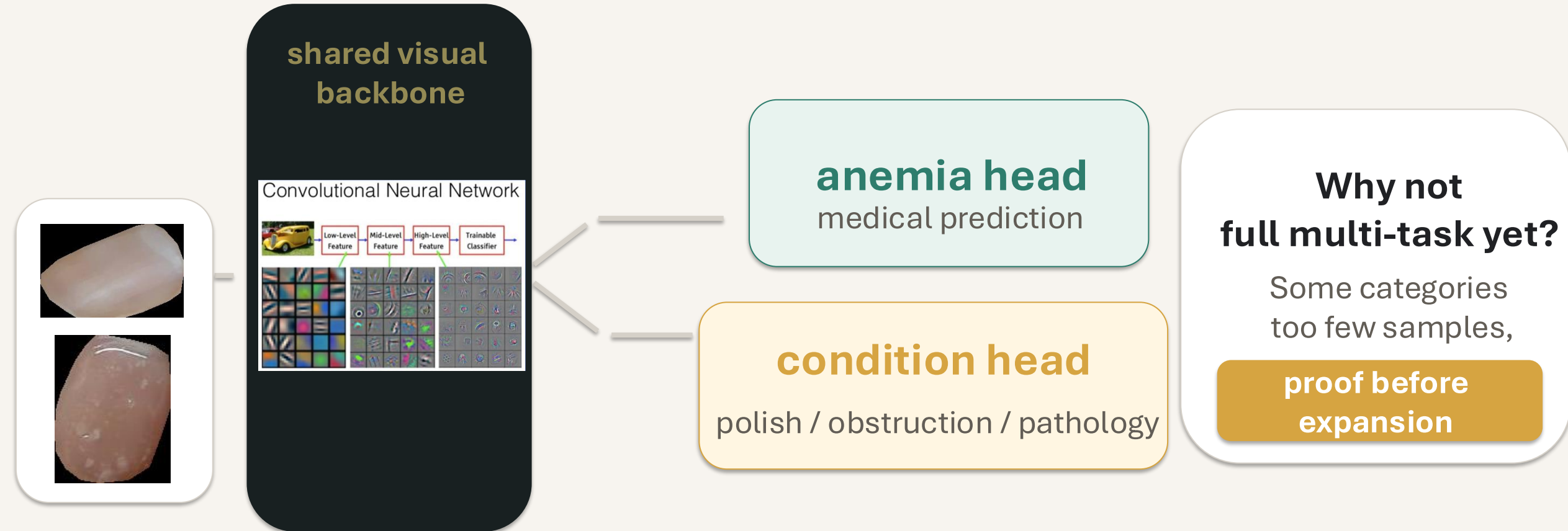


Interpretation

no-polish recall
0.98

polish recall
0.82

The binary classifier is a safe first step toward a multi-task condition head.



The main lesson is that robustness starts before the main model.

What worked

Segmentation made the visual cue clearer

Group split made evaluation more honest.

What is limited

The task is still binary, imbalanced, and not yet a clinical anemia result.

What comes next

Add rare-condition data carefully

Test whether condition awareness improves real anemia robustness.

Final takeaway: “unusable” data is only useless if we never ask what kind of signal it contains.

Q&A